

GENERAL THETA FOUNDATIONS I: FINITE ALPHABETS, SPECTRAL GAPS, AND HIDDEN MEMORY

QIAN QI

ABSTRACT. We study the width of finite stochastic transducers that reproduce every conditional probability of a prescribed orthogonal command experiment. For a fixed finite alphabet whose action on the sphere has an L^2 spectral gap, we prove lower and upper widths of orders $(N/\log(N+1))^{(d-1)/2}$ and $N^{(d-1)/2}$, respectively. The upper realization is exact; the lower bound permits arbitrary hidden states, unrelated transition rows at different epochs, and fixed small output error. Five explicit rational three-dimensional commands therefore have classical width between $cN/\log(N+1)$ and CN , although a single qubit realizes the same probabilities. This probability task is distinct from strict-cutpoint language equivalence. A second finite, noncommuting class, consisting of permutations and conjugations of resonant planar modes, has the matched order $N^{1/3}$. Its full group need not possess an irrational one-dimensional character. The common argument is a packet compression inequality formulated for the entire fresh command word, without assuming that its product is sufficient for the hidden machine. We also give a finite fair-bit compiler for sparse rows, with logarithmic rather than denominator-sized control overhead. The principal width is nonuniform clocked atomic-row width; the fair-bit interface is separately self-paced. The spectral-gap bounds determine a polynomial exponent, not a matched order up to constants.

1. INTRODUCTION

A small positive factorization at every observation cut does not, by itself, produce a small machine that realizes all cuts simultaneously. The additional requirement is causal compatibility: the same retained state must support each possible next input. We investigate this requirement for command experiments

$$(1) \quad \mathbb{P}(B = b \mid x, a_1 \cdots a_N, j) = \frac{1 + b\rho e_j^T U_{a_N} \cdots U_{a_1} x}{2}, \quad b \in \{-1, 1\}.$$

The command alphabet is fixed and finite. The seed is supplied once, and one coordinate query is supplied after the prescribed horizon. The command matrices act orthogonally; the realizing matrices act stochastically and may have arbitrary hidden directions.

The finite-alphabet restriction is substantive. An independent Haar rotation cannot be treated as one input when only a finite gate set is available. Nor can a whole word be replaced by its group product in the hidden transition law: equal products can have different intermediate computations. The present results retain each input slot and each full word throughout the converse.

Let $m = d - 1$. Suppose a probability distribution on the fixed alphabet induces an averaging operator P with

$$(2) \quad \|Pf\|_{L^2(\sigma)} \leq \lambda \|f\|_{L^2(\sigma)} \quad \text{when} \quad \int f d\sigma = 0, \quad \lambda < 1,$$

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where σ is normalized spherical measure. For fixed positive signal and sufficiently small uniform error, Theorem 4.3 proves

$$(3) \quad c \left(\frac{N}{\log(N+1)} \right)^{m/2} \leq W_{N,\varepsilon} \leq W_{N,0} \leq CN^{m/2}.$$

In particular $\log W_{N,\varepsilon} / \log N \rightarrow m/2$. The logarithmic gap in (3) is part of the statement, not absorbed into a Θ notation. The proof uses only a finite command distribution. A logarithmic number of actual commands makes its orbit distribution sufficiently close to spherical measure at the resolution imposed by the endpoint width.

Theorem 5.1 instantiates this criterion with five rational matrices in $\text{SO}(3)$. The spectral gap is an external theorem of Bourgain–Gamburd [1], not an assumption checked by numerics. Density and the algebraic hypotheses are verified for the stated gates. The experiment has separate positive minimum four at every internal cut, but its compatible hidden width grows almost linearly. A two-dimensional quantum memory realizes it exactly. Proposition 5.2 makes this comparison operational rather than terminological.

A different geometry gives a different order. In Theorem 6.1, finitely many orthogonal monomial commands permute planar modes, conjugate them, and rotate them by integer multiples of one badly approximable angle. Their exact and fixed-error widths are $\Theta(N^{1/3})$. The group generated by the displayed five-letter example is noncommuting, its real representation is irreducible, and no continuous one-dimensional character sees its irrational rotations. This is not an assertion that every finite noncommuting alphabet has the same exponent.

The methodological distinction is between mixing of the uncompressed orbit and irreversible loss at a finite register. The former supplies an orbit quantization scale; the latter converts that scale into a cumulative width constraint. All lower bounds allow new machines at each horizon and arbitrary changes of stochastic rows between epochs. The exact upper machines have common command rows for a fixed horizon, with a horizon-dependent decoder.

Notation	Meaning
K_t, W	available labels at cut t , and their maximum
$W_{N,\varepsilon}$	least peak at uniform binary row-TV error ε
$W, g(W)$ in §3	fresh packet word, and its ordered group product
$\Gamma_k(\mu)$	least k -center orbit distortion, including the starting direction
B_k	number of actual commands used by a test packet
q_t	norm of the conditional orbit mean, averaged over hidden states

The use of W for a word is local to the packet lemma; width elsewhere is scalar. Classical positive realization, spherical covering, spectral gap, and strict-cutpoint simulation are identified at their points of use. The new finite-alphabet consequences depend on these stated inputs. The sharp binary rotation result of [10] is retained and its packet and resonant arguments are reproduced within the larger monomial construction. Historical results not needed for the present proofs remain in the separately preserved supporting article.

2. FINITE EXPERIMENTS AND PERSISTENT REGISTERS

Fix a finite alphabet $\mathcal{A}$, matrices $U_a \in O(d)$, a finite seed set $X \subset \mathbb{S}^{d-1}$ and $0 < \rho \leq 1/10$. Unless otherwise stated, $X = \{\pm e_1, \dots, \pm e_d\}$ and $\text{Id} \in \mathcal{A}$. We use the full specification (1); no distribution or fairness assumption is imposed on the external word.

Definition 2.1 (Nonuniform clocked stochastic transducer). For horizon N , a transducer consists of finite state sets $[K_t]$, initial distributions E_x , stochastic matrices $T_{t,a} \in \mathbb{R}_+^{K_t \times K_{t+1}}$, and terminal columns $d_j \in [-1, 1]^{K_N}$. Its mean on a word is $E_x T_{0,a_1} \cdots T_{N-1,a_N} d_j$. Its error is half the

maximum absolute discrepancy from the mean in (1). The peak counts the available labels $\max_t K_t$, and also two terminal labels when the emitted answer is retained at a charged cut. The least peak with error at most ε is $W_{N,\varepsilon}$.

Only the updated register persists. The seed and the query appear once; the current command is consumed in its update. Private randomness does not persist outside the register. An input-dependent clock, a previous command, or an installation tag would be charged if retained. The epoch, N , complete row tables, their construction and arithmetic, and exact atomic sampling are free. Rows may vary with t and N . Unused rows are normalized, and padding counts as available labels. This is maximum layer width of a nonuniform probabilistic ordered read-once program, not uniform program space or total branching-program size. Section 7 specifies a different, finite-coin interface.

For comparison with the earlier compact-input theorem, a compact command space requires each matrix entry $a \mapsto T_{t,a}(s, s')$ to be Borel. Such a row function is nonuniform program data, not a finitely represented real-input algorithm. None of the finite-alphabet theorems below requires that access. Every input to their machines is a symbol from a fixed finite set.

Proposition 2.2 (Separate positive minimum). *If $\rho d < 1$ and the identity is available, the normalized residual rank and the separate positive minimum at every cut are $d + 1$. A separate minimum places no cap on the other registers.*

Proof. Residual rows are affine in $v = \rho U_w x$. Idle prefixes from $\pm e_j$ span $\mathbb{R}^d$, and an idle suffix followed by the coordinate queries separates the coordinates. Normalization adds the constant row, giving rank $d + 1$. A regular simplex centered at zero with circumradius $d\rho$ has inradius ρ . It contains every reachable vector and all its vertices have norm less than one. Each vertex consequently defines a valid future binary channel after every orthogonal command word. At the selected cut decompose v into these vertices. Before this cut one may store the raw prefix; afterwards one may store a simplex label and the raw suffix. At l later inputs this costs at most $(d + 1)|A|^l$ labels, and remains finite. It embeds the factorization in a complete machine rather than an amputated static matrix. Rank gives the converse. $\square$

Lemma 2.3 (Terminal correlation). *For a fixed unit seed x , put $Y_t = U_{a_t} \cdots U_{a_1} x$ and $q_t = \mathbb{E}[\|Y_t\| \mid S_t]$ under any external word law independent of the machine's private randomness. If the wordwise error is at most ε , then*

$$(4) \quad q_N \geq \kappa_d := \rho/\sqrt{d} - 2\varepsilon.$$

Proof. Combine the d decoder columns into $D(S_N) \in [-1, 1]^d$. Conditionally on a word, its mean differs from ρY_N by a vector of norm at most $2\sqrt{d}\varepsilon$. Thus $\mathbb{E}\langle Y_N, D(S_N) \rangle \geq \rho - 2\sqrt{d}\varepsilon$. Conditional expectation bounds the absolute value of this quantity by $\sqrt{d}q_N$. Two or more queries are not jointly sampled in this argument; their decoder columns are only combined in the proof. $\square$

3. COMPRESSION OF FULL COMMAND WORDS

Let a compact group act orthogonally on a finite-dimensional real Hilbert space H . Complex unitary representations are covered by the real inner product. For a group probability measure μ , define

$$(5) \quad \Gamma_k(\mu) = \inf_{v, u_1, \dots, u_k \in \mathbb{S}(H)} \int \min_{i \leq k} \{1 - \langle U_g v, u_i \rangle\} d\mu(g).$$

This continuous minimization has a minimum and lies in $[0, 1]$. The upper bound follows by including the direction of $\int U_g v d\mu(g)$ as a center. An invariant direction may make the coefficient zero. The conclusions below remain valid but then give no positive loss from that packet.

Lemma 3.1 (Full-word packet contraction). *Let $Y \in H$ be integrable and S finite. Let a random packet word W be independent of $\sigma(Y, S)$, and let $g(W)$ be its measurable ordered product. Suppose S' has at most k positive-mass values and, for every label r ,*

$$(6) \quad \mathbb{P}(S' = r \mid \sigma(Y, S, W)) = T_W(S, r) \quad a.s.,$$

where each T_w is a stochastic kernel. If $\mu = g_* \text{Law}(W)$, then

$$(7) \quad \mathbb{E} \|\mathbb{E}[U_{g(W)} Y \mid S']\| \leq (1 - \Gamma_k(\mu)) \mathbb{E} \|\mathbb{E}[Y \mid S]\|.$$

In particular T_w need not be constant on fibers of g .

Proof. Write $p_s = \mathbb{P}(S = s)$, $z_s = \mathbb{E}[Y \mid S = s]$, and let ν be the law of W . Independence and (6) imply

$$p'_r z'_r = \sum_s p_s \int T_w(s, r) U_{g(w)} z_s d\nu(w), \quad z'_r = \mathbb{E}[U_{g(W)} Y \mid S' = r].$$

For nonzero z'_r take $u_r = z'_r / \|z'_r\|$, with arbitrary unit choices for zero centroids. Summing inner products gives

$$\begin{aligned} \sum_r p'_r \|z'_r\| &= \sum_s p_s \int \sum_r T_w(s, r) \langle U_{g(w)} z_s, u_r \rangle d\nu(w) \\ &\leq \sum_{s: z_s \neq 0} p_s \|z_s\| \int \max_r \langle U_{g(w)} z_s / \|z_s\|, u_r \rangle d\nu(w) \\ &\leq (1 - \Gamma_k(\mu)) \sum_s p_s \|z_s\|. \end{aligned}$$

Only after removing the word-dependent assignment by a pointwise maximum do we push ν forward to μ . No product-sufficiency assertion is used. The old register and all internal packet cuts are unrestricted. $\square$

Remark 3.2 (Why the word matters). In a group with an involution r , the words ee and rr have the same product. A transducer can nevertheless record which first symbol occurred. Thus integration of the internal computation gives a kernel T_w , not necessarily a kernel $T_{g(w)}$. Also, independence of the product alone would be insufficient: an identity-product word correlated with Y could reveal Y through its spelling. The hypothesis concerns the entire fresh word. This repairs the word/product mismatch in the preceding formulation without restricting the class of machines or packet supports.

Theorem 3.3 (Packet budget). *Start with a fixed unit vector. On disjoint intervals use packet words whose full laws are fixed in advance, each independent of the earlier process and of all private random choices. Fixed commands may fill the gaps. If k_i bounds the register at packet endpoint i , and a terminal correlation forces $q_N \geq \kappa > 0$, then*

$$(8) \quad \kappa \leq \prod_i (1 - \Gamma_{k_i}(\mu_i)), \quad \sum_i \Gamma_{k_i}(\mu_i) \leq \log(1/\kappa).$$

The full words and their products may be noncommuting, and the product map may have collisions.

Proof. The ordered internal stochastic matrices of a whole packet multiply to T_w . Their Markov property gives (6); word freshness gives the required independence. Apply Lemma 3.1 at successive boundaries. A fixed command followed by stochastic compression cannot increase q , by conditional Jensen. Initially $q = 1$. Multiplication proves the product, and $1 - u \leq e^{-u}$ proves the sum. A factor zero precludes positive terminal correlation, as asserted. $\square$

For a wordwise specification the test law may be chosen after the available-label profile is known. It does not become information given to the machine. In the next section packets are made of independent letters. In Section 6 they have correlated letters; both satisfy the same full-word hypothesis.

4. FINITE SPECTRAL-GAP ALPHABETS

Let $d \geq 2$, $m = d - 1$, and σ be normalized measure on $\mathbb{S}^{d-1}$. For a distribution p on the finite alphabet, put $Pf(v) = \sum_a p(a)f(U_a v)$. Our hypothesis is the absolute L^2 gap

$$(9) \quad \|P|_{L^2_0(\sigma)}\| \leq \lambda < 1, \quad 0 < \lambda < 1.$$

Laziness makes an upper spectral gap absolute for symmetric walks. We assume (9) on the full mean-zero function space, not only on the defining d -dimensional representation. Density alone is not substituted for this analytic hypothesis.

Lemma 4.1 (Uniform mixing for Lipschitz tests). *There is C_d such that every real 1-Lipschitz f with values in $[-1, 1]$ satisfies*

$$(10) \quad \sup_v |P^B f(v) - \sigma f| \leq C_d \lambda^{2B/(m+2)}.$$

For $d = 3$ one can take $C_3 = 3$.

Proof. Orthogonal actions preserve distances, so $h = P^B f - \sigma f$ is 1-Lipschitz. Its L^2 norm is at most λ^B , since $\|f - \sigma f\|_2^2$ is a variance bounded by one. A chordal radius- r ball has mass at least $c_d r^m$ for $0 < r \leq 1$. Averaging $|h(v)| \leq |h(u)| + r$ over this ball and using Cauchy–Schwarz gives

$$|h(v)| \leq r + c_d^{-1/2} r^{-m/2} \lambda^B.$$

Take $r = \lambda^{2B/(m+2)}$. The ball estimate follows from polar integration and $\sin t \geq 2t/\pi$ on $[0, \pi/2]$. On $\mathbb{S}^2$ the chordal cap mass is exactly $r^2/4$; taking $r = \lambda^{B/2}$ gives $3\lambda^{B/2}$. $\square$

Lemma 4.2 (Finite-word orbit distortion). *There are constants $c_d > 0$ and integers $B_k = O_{d,\lambda}(\log(k+1))$ such that the product law μ_{B_k} of B_k independent command letters satisfies*

$$(11) \quad \Gamma_k(\mu_{B_k}) \geq c_d k^{-2/m}.$$

For $d = 3$, the explicit choices

$$(12) \quad B_k = \left\lceil \frac{2 \log(12k)}{-\log \lambda} \right\rceil, \quad \Gamma_k(\mu_{B_k}) \geq \frac{1}{4k}$$

are valid.

Proof. A chordal radius- r cap has mass at most $A_d r^m$, with $A_d \geq 1$. For any k unit centers, choose $r = (2A_d k)^{-1/m}$. Outside their union, of mass at least one half, the loss $1 - \max_i \langle v, u_i \rangle$ is at least $r^2/2$. Consequently the spherical average loss is at least $a_d k^{-2/m}$, where $a_d = (2A_d)^{-2/m}/4$. The function $f(v) = \max_i \langle v, u_i \rangle$ is 1-Lipschitz and lies in $[-1, 1]$. Lemma 4.1 gives, uniformly in the centers and starting direction,

$$P^B f(v) \leq 1 - a_d k^{-2/m} + C_d \lambda^{2B/(m+2)}.$$

Choose B_k so the last term is at most $a_d k^{-2/m}/2$. This proves the first assertion. Powers of P and chronologically ordered products give the same law here, since the independent letters have a common distribution and their joint law is invariant under reversal.

On $\mathbb{S}^2$ take $r = \sqrt{2/k} \leq \sqrt{2}$. The exact cap mass $r^2/4$ gives spherical loss at least $1/(2k)$. Equation (12) makes $3\lambda^{B_k/2} \leq 1/(4k)$, proving the sharper constants. Notice that B_k counts ordinary inputs, not one Haar input. $\square$

Theorem 4.3 (Finite-alphabet width and occupation). *Assume (9), $0 < \rho \leq 1/10$, and $0 \leq \varepsilon < \rho/(2\sqrt{d})$. Then*

$$(13) \quad c_{d,\lambda,\rho,\varepsilon} \left(\frac{N}{\log(N+1)} \right)^{m/2} \leq W_{N,\varepsilon} \leq W_{N,0} \leq C_d N^{m/2}.$$

In particular $\lim_{N \rightarrow \infty} \log W_{N,\varepsilon} / \log N = m/2$. Every available-label profile with this error satisfies

$$(14) \quad \#\{1 \leq t \leq N : K_t \leq k\} \leq B_k - 1 + c_d^{-1} B_k k^{2/m} \log(1/\kappa_d).$$

The other cuts may have arbitrarily large registers. For $d = 3$ one may replace the second term by $4B_k k \log(1/\kappa_3)$ using (12). The exact upper realization has common command rows and an N -dependent decoder.

Proof of the lower and occupation bounds. Under the independent-letter law, every consecutive length- B_k block is a fresh packet. Lemma 2.3 and Theorem 3.3 imply for a peak- k machine

$$(15) \quad \lfloor N/B_k \rfloor c_d k^{-2/m} \leq \log(1/\kappa_d).$$

Thus $N < B_k(1 + c_d^{-1} k^{2/m} \log(1/\kappa_d))$. As $B_k \leq C \log(k+1)$, this implies the first inequality in (13). Explicitly, if $k \geq N^{m/2}$ it is immediate; otherwise $\log(k+1) \leq C_m \log(N+1)$ and rearrangement suffices. Constants absorb the finitely many small N .

For occupation, discard at most $B_k - 1$ early endpoints and greedily select low-width endpoints at spacing at least B_k . Each selected interval removes at most B_k candidate endpoints. If l intervals are selected, their total loss is at least $l c_d k^{-2/m}$, irrespective of the internal widths. The packet budget bounds this by $\log(1/\kappa_d)$ and proves (14). The initialization cut is not in this count. The argument remains valid for positive-mass states under the selected independent-letter law. $\square$

Lemma 4.4 (Rational spherical synthesis). *For every $N \geq 1$ there is a rational finite set $V_N \subset \mathbb{S}^{d-1}$ of cardinality at most $C_d N^{m/2}$ such that*

$$\eta_N \mathbb{B}^d \subset \text{conv } V_N, \quad \eta_N = 1 - \frac{1}{8N}.$$

For finitely many orthogonal commands, this gives the exact upper bound in (13). If the commands, seeds and ρ are rational, all transition, initialization and decoder entries may be chosen rational, with at most $d+1$ successors in each command row.

Proof. Use the two stereographic charts

$$t \mapsto \left(\frac{2t}{1 + \|t\|^2}, \pm \frac{1 - \|t\|^2}{1 + \|t\|^2} \right), \quad t \in [-1, 1]^m.$$

Every sphere point has a preimage in the unit ball in one chart. Each chart is 2-Lipschitz. A grid of mesh $1/M$, with $M = \lceil 2\sqrt{mN} \rceil$, is therefore within chordal distance $\sqrt{m}/M$ of every sphere point. Its at most $2(2M+1)^m$ images are rational. In every unit direction the support function of their convex hull is at least $1 - m/(2M^2) \geq \eta_N$.

Write $P = \text{conv } V_N$, $a_t = \eta_N^{-t}/4$. From label $u \in V_N$ on command a , choose coefficients expressing $\eta_N U_a u$ as a convex combination of V_N . This is possible by ball containment. The coefficients depend on a, u, N , not on the epoch. Initialize to mean ρx in $a_0 P$, since $\rho \leq 1/10 < 7/32 \leq \eta_N/4$. At the final cut use mean decoder $a_N e_j^T u$. Bernoulli's inequality gives $\eta_N^N \geq 7/8$, so $a_N \leq 2/7 < 1$. The conditional mean update is $a_{t+1} \mathbb{E}[u' | u, a] = a_t U_a u$, proving exactness by induction. Only the vertex label is retained.

For rational data, convex feasibility is a rational linear system. A basic feasible solution has at most $d+1$ nonzero coefficients and is rational; apply this separately to the finitely many rows and

seeds. An extra binary-output cut has two labels and does not change the bound. This proves the upper assertion of Theorem 4.3. $\square$

The mixing and quantization inequalities are classical analytic and geometric mechanisms. Their use above is a finite-word, all-hidden-state compression bound. A logarithmic mixing time is explicitly paid. Neither an atomic Haar command nor a uniform-in- d constant is used.

5. FIVE RATIONAL COMMANDS AND A QUBIT

Consider the fixed matrices

$$(16) \quad R_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -7/25 & -24/25 \\ 0 & 24/25 & -7/25 \end{pmatrix}, \quad R_z = \begin{pmatrix} -7/25 & -24/25 & 0 \\ 24/25 & -7/25 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

and the five-letter alphabet $\mathcal{A}_* = \{\text{Id}, R_x, R_x^{-1}, R_z, R_z^{-1}\}$. Use the six seeds $\{\pm e_j : 1 \leq j \leq 3\}$, signal $\rho = 1/10$, and three terminal coordinate queries. Every specified probability lies in $[9/20, 11/20]$ at every horizon. Both the alphabet and the matrices are independent of N .

Theorem 5.1 (A finite noncommuting rational family). *For the experiment (16), every fixed $0 \leq \varepsilon < 1/(20\sqrt{3})$ has*

$$(17) \quad c_\varepsilon \frac{N}{\log(N+1)} \leq W_{N,\varepsilon}(\mathcal{A}_*) \leq W_{N,0}(\mathcal{A}_*) \leq CN.$$

The exact upper machine can have rational entries and common command rows. Every separate positive minimum is four. If $\lambda < 1$ is the sphere gap constant for the lazy walk with weights $1/2, 1/8, 1/8, 1/8, 1/8$, then a peak- K machine necessarily obeys

$$(18) \quad N < B_K \{1 + 4K \log(1/\kappa_3)\}, \quad B_K = \left\lceil \frac{2 \log(12K)}{-\log \lambda} \right\rceil, \quad \kappa_3 = \frac{1}{10\sqrt{3}} - 2\varepsilon.$$

No numerical value of the spectral gap is asserted.

Proof. Let $\sigma_x, \sigma_y, \sigma_z$ denote the Pauli matrices and put $A_x = (3I - 4i\sigma_x)/5$, $A_z = (3I - 4i\sigma_z)/5$ in $\text{SU}(2)$. Their adjoint actions on the traceless Hermitian matrices are precisely R_x, R_z . Their eigenvalues include $(3 + 4i)/5$. This number is not a root of unity: otherwise its sum with its inverse, $6/5$, would be a rational algebraic integer, hence an integer. Their powers are therefore dense in the respective one-parameter axial subgroups. The Lie algebra generated by $i\sigma_x, i\sigma_z$ contains their bracket, a nonzero multiple of $i\sigma_y$, so the closed subgroup they generate is all of $\text{SU}(2)$. In particular the generators do not commute.

The matrices have algebraic entries. Bourgain–Gamburd’s spectral-gap theorem [1, Theorem 1] applies to their symmetric averaging operator on $L^2(\text{SU}(2))$. Adding probability $1/2$ at the identity makes its spectrum nonnegative and its mean-zero norm strictly below one. The action on the sphere is a homogeneous quotient: lifting $f(v)$ to $f(gv_0)$ embeds $L^2(\mathbb{S}^2)$ isometrically into $L^2(\text{SU}(2))$ and intertwines the averaging operators. Hence the same gap holds on the sphere.

Apply Theorem 4.3, with the explicit constants in (12). Rational synthesis follows from Lemma 4.4. Proposition 2.2 gives four at each separate cut, since $3\rho < 1$. The spectral-gap theorem is the substantive external input; no finite collection of eigenvalue computations is used as its proof. $\square$

This theorem has a different arithmetic content from the binary example with rational rotation entries. One rotation may have exceptionally slow circle recurrences. Two noncommuting algebraic axes instead invoke a spectral gap on the entire sphere. We do not infer that either individual rotation angle is badly approximable.

Proposition 5.2 (Quantum probabilities versus threshold languages). *The experiment in Theorem 5.1 has an exact single-qubit realization with a finite, horizon-independent collection of operations. A general one-way quantum automaton of dimension two realizes its probabilities on valid prepare–command–query words. For any fixed strict cutpoint, its threshold language has a PFA realization with at most five states by [2, Theorem 3.2]. Nevertheless any classical clocked transducer approximating these numerical probabilities at the stated tolerance has width bounded below by (17).*

Proof. For seed x prepare

$$\varrho_x = \frac{1}{2}(I + \rho(x_1\sigma_x + x_2\sigma_y + x_3\sigma_z)).$$

It is a density matrix because $\|x\| = 1$ and $\rho < 1$. A command uses the corresponding unitary $I, A_x^{\pm 1}, A_z^{\pm 1}$. Their conjugations rotate the Bloch vector by (16). Measuring σ_j yields $\text{tr}(\varrho_w(I + b\sigma_j)/2) = (1 + b\rho e_j^T U_w x)/2$, exactly the required law. A query-dependent unitary before a fixed σ_z measurement implements all three queries; a seed symbol is a completely positive reset channel. Thus the operations define a dimension-two general quantum automaton on a finite alphabet, including all words outside the promised protocol with its usual automaton semantics.

The cited strict-cutpoint conversion concerns that entire language and therefore also its valid words. It preserves threshold signs, not numerical probabilities. In the conversion, the deviation from the classical cutpoint is multiplied by a positive length-dependent shrinking factor [2, Proposition 3.1]. Such attenuation is harmless for language equivalence and cannot supply the fixed-correlation requirement of Lemma 2.3. The last assertion is exactly Theorem 5.1; it even permits horizon-dependent and nonstationary classical simulators. $\square$

The variable in (17) is input length, with quantum dimension fixed. The quadratic state-cost results of Chen–Wu [2, 3] instead vary quantum dimension under strict-cutpoint equivalence. Neither result is contradicted or strengthened in its own metric by changing to the other objective.

6. A MATCHED NONCOMMUTING FINITE CLASS

The preceding finite spectral-gap class determines a growth exponent, with a logarithmic order gap. There is also a finite noncommuting class with a matched order. Its orbit structure is different from a sphere.

Identify $\mathbb{R}^{2r}$ with $\mathbb{C}^r$, fix an irrational α , and put $\zeta = e^{2\pi i\alpha}$. A *resonant monomial command* has the form

$$(19) \quad (U_a z)_j = \zeta^{n_{a,j}} \begin{cases} z_{\pi_a^{-1}(j)}, & \epsilon_{a,j} = 1, \\ \overline{z_{\pi_a^{-1}(j)}}, & \epsilon_{a,j} = -1, \end{cases}$$

where π_a is a permutation, $n_{a,j}$ is an integer, and $\epsilon_{a,j} \in \{-1, 1\}$. Fix finitely many such commands, and let $M = \max(1, \max_{a,j} |n_{a,j}|)$. Include the identity and the command A that multiplies the first coordinate by ζ and fixes the rest. Take the $4r$ seeds $(\pm 1 \pm i)e_j$, signal $0 < \rho \leq 1/(10r)$, and one of the $2r$ real coordinate queries. The specified mean is the corresponding coordinate of $\rho U_w x$.

Theorem 6.1 (Sharp monomial width). *If $b_\alpha := \inf_{l \geq 1} l \|l\alpha\|_{\mathbb{R}/\mathbb{Z}} > 0$, then for each fixed $0 \leq \varepsilon < \rho/2$,*

$$(20) \quad W_{N,\varepsilon}^{\text{mon}} = \Theta_{r,A,\alpha,\rho,\varepsilon}(N^{1/3}).$$

The exact upper machine has common command rows and at most four successors in any such row. Its final decoder may depend on N . Every separate exact positive minimum is $2r + 1$.

Lower bound. Restrict to seed $(1+i)e_1$, commands Id, A , and the first two real queries. The conditional target is the planar phase $\rho(1+i)\zeta^{L_N}$. Combining its two mean columns shows that $q_N \geq \kappa = \rho - 2\varepsilon$, as in Lemma 2.3 with the sharper planar normalization.

For a proposed peak k , put $B = 2k - 1$ and choose the length- B word $A^J \text{Id}^{B-J}$ with J uniform on $\{0, \dots, B\}$. Its $2k$ circle points are separated by $s_B = \min_{1 \leq l \leq B} \|l\alpha\|$. Each open arc of radius $s_B/2$ contains at most one of these points. Any k centers therefore leave at least half of them at distance at least $s_B/2$. The average loss is at least $\frac{1}{2}(1 - \cos(\pi s_B)) = \sin^2(\pi s_B/2) \geq s_B^2$. The full-word lemma applies although the packet's letters are correlated. Consecutive independent packets imply

$$\lfloor N/B \rfloor \frac{b_\alpha^2}{B^2} \leq \log(1/\kappa), \quad N < B + b_\alpha^{-2} B^3 \log(1/\kappa).$$

This proves the lower order against every hidden realization. No character of the full command group was used, only a sublanguage. $\square$

Sparse exact upper construction. Take a convergent p_0/q_0 of α , put $q = 2q_0$ and $\delta = 2\pi(\alpha - p_0/q_0)$. Choose the first convergent for which

$$(21) \quad q \geq \max\{8M, 8, (160MN)^{1/3}\}.$$

Then $|\delta| < 8\pi/q^2$ and $|n\delta| < \pi/q$ for $|n| \leq M$. Set $a = \pi/q$, and for each needed integer n put

$$\lambda_n = \frac{\cos(a - |n\delta|)}{\cos a}, \quad \omega_n = \frac{\sin |n\delta|}{\lambda_n \sin(2a)}, \quad \Lambda = \max_{|n| \leq M} \lambda_n.$$

These numbers obey $1 \leq \lambda_n \leq \Lambda$ and $0 \leq \omega_n \leq 1$. Let $u_l = e^{2\pi i l/q}$, indices modulo q . The two-point law on labels $l + 2np_0$ and $l + 2np_0 + \text{sgn}(n\delta)$, of weights $1 - \omega_n, \omega_n$, has mean $\zeta^n u_l / \lambda_n$. For $n = 0$ interpret it as the single label l . This is the intersection of a polygon edge with the rotated ray.

The register is a pair (j, l) , comprising one mode and one polygon label. Command a sends the mode to $j' = \pi_a(j)$ and, if needed, replaces l by $-l$ to conjugate. Draw the just described edge label for $n = n_{a,j'}$. With probability $(1 + \lambda_n/\Lambda)/2$ retain it; otherwise replace it by its antipode, label $l + q/2$. The expected complex label is exactly the commanded one divided by Λ . At most four labels occur. This antipodal step prices a common radial factor without a free zero state or a dense uniform redraw.

Associate $a_t u_l e_j$ with label (j, l) , where $a_t = \Lambda^t/4$. The conditional mean update is exactly U_a . Moreover

$$\log \Lambda \leq M|\delta| \tan a \leq 16\pi^2 M/q^3 < 160M/q^3, \quad a_N \leq e/4 < 1.$$

The seed occupies one mode. Its planar vector has radius $\rho\sqrt{2} \leq \sqrt{2}/10 < \cos(\pi/8)/4$, hence lies in the initial polygon and can be initialized using at most three vertices. Decoder means are the real coordinates of $a_N u_l e_j$, with zeros on other modes. Induction gives the exact specified law on all words, not just the restricted lower-bound words. Only (j, l) persists.

Bad approximability is equivalent to bounded partial quotients, which bound the ratio of consecutive convergent denominators. Thus the first choice in (21) has $q = O_{\alpha, M}(N^{1/3})$, proving the upper order with rq labels. $\square$

Separate minima. The seed vectors span $\mathbb{R}^{2r}$; identity suffixes separate their coordinates, so normalized residual rank is $2r + 1$. All predictive vectors have norm $\rho\sqrt{2}$. A regular simplex with inradius $\rho\sqrt{2}$ has circumradius $2r\rho\sqrt{2} \leq \sqrt{2}/5 < 1$. Its vertices define legal continuations under every orthogonal command. The selected-cut construction of Proposition 2.2 gives the matching positive minimum, with finite raw-word storage elsewhere. $\square$

Proposition 6.2 (A full group not seen by a character). *For every $r \geq 2$, at most five commands suffice to generate an instance of Theorem 6.1 whose compact closure is $O(2)^r \rtimes S_r$. Its real representation on $\mathbb{R}^{2r}$ is irreducible. Every continuous one-dimensional unitary character is trivial on the connected torus $SO(2)^r$, hence cannot see A 's irrational phase.*

Proof. Use Id, A , conjugation of the first mode, the cyclic permutation of modes, and the transposition of the first two modes. The last two permutations generate S_r ; for $r = 2$ duplicate commands are removed. Irrational powers of A fill one circle, and permutation conjugates fill all coordinate circles. Conjugations supply their reflections, proving the asserted closure.

An invariant real subspace under these circle actions decomposes into coordinate planes: averaging over all other circles projects onto a chosen plane. A nonzero projection contains that whole plane by rotation, and permutation invariance then supplies all planes. This proves irreducibility. Finally, conjugation of a coordinate circle element sends it to its inverse. In a one-dimensional character its value must therefore equal its inverse. Continuity on the connected circle forces that value to be one. Apply this in each coordinate. $\square$

This example is genuinely noncommuting and is not an application of a character of its entire compact group. Its lower bound still uses a cyclic sublanguage, and its upper bound uses the monomial structure. It does not classify arbitrary finite dense subsets of $SO(d)$. The case $r = 1$, $\rho = 1/10$ and commands Id, A recovers the original sharp rotation experiment of [10] without changing its signal, seed set, error tolerance or hidden-state quantifier.

7. FINITE COINS WITH CHARGED CONTROL

The atomic model permits arbitrary exact row probabilities. We now separate a finite random-bit implementation. The environment supplies an input only at a ready boundary and waits until the update has finished. Fair bits are private. The row table may contain precomputed finite binary thresholds, but the bit position, current input, old label and control state are all charged. There is no free microstep clock.

Theorem 7.1 (Sparse-row dyadic compiler). *Suppose a finite command transducer has peak K , at most L successors per row, fixed finite seed and query alphabets, and common command rows. For any $\delta > 0$ it has a self-paced fair-bit implementation whose uniform terminal row-TV error increases by at most δ , using*

$$(22) \quad C_{A,X,J,L}K(b+1) \text{ persistent labels,} \quad b = \max \left\{ 1, \left\lceil \log_2 \frac{C_L(N+2)}{\delta} \right\rceil \right\},$$

and at most $(L-1)b$ fair bits for a command update. The same bound holds for sparse initialization and the binary terminal decoder. Tables and the prescribed macroscopic final boundary remain nonuniform.

Proof. A categorical row with at most L outcomes is a binary decision tree with at most $L - 1$ internal nodes. At each node express the probability of choosing one subtree conditionally on reaching that node. A zero-mass node is unreachable and may use an arbitrary conditional row. Round each conditional probability down to a multiple of 2^{-b} . Coupling the binary decisions shows that the row-TV discrepancy is at most $(L - 1)2^{-b}$, without needing the original probabilities to be rational. Initialization and binary decoding are treated in the same way.

Thresholds zero and one are deterministic. For $0 < m < 2^b$, to sample the dyadic Bernoulli threshold $m/2^b$, compare independent fair bits with the b binary digits of m , from most to least significant. At the first mismatch the comparison is decided; if all digits agree, return the prescribed strict-comparison outcome. No prefix integer is stored. The comparison retains only its position $i \in \{1, \dots, b\}$; while all bits agree, their values are already the fixed threshold prefix.

A categorical tree additionally retains one of at most $L - 1$ nodes. It must retain the old label and the current input until a leaf gives the new label. These are all included in (22), along with ready, seed and query phases. There is no second uncharged output index. The common-row assumption eliminates a retained epoch counter inside the sampler. An external end query selects the final decoder after the prescribed number of completed inputs.

TV contraction under stochastic kernels and induction on the actual word give at most $C_L(N + 2)2^{-b}$ terminal discrepancy. This is at most δ by the chosen b . The argument couples legal completed updates and therefore prices every command even when the bit comparison stops early. Its duration is at most $(L - 1)b$ bits per update. $\square$

Corollary 7.2 (Leading label bits with finite coins). *For every fixed positive error below the thresholds of the preceding theorems, their explicit machines admit the following finite-coin peaks:*

$$\begin{aligned} O(N^{1/3} \log(N + 1)) & \quad \text{for the resonant monomial class,} \\ O(N^{(d-1)/2} \log(N + 1)) & \quad \text{for the spectral-gap class.} \end{aligned}$$

Consequently the least persistent bits in this self-paced, nonuniform fair-bit model have leading coefficients $1/3$ and $(d - 1)/2$ in $\log_2 N$, respectively, with additive $O(\log \log(N + 1))$ uncertainty.

Proof. The monomial upper has at most four successors; the spherical upper has at most $d + 1$. Their initialization is sparse and their decoder binary. Compile the exact machine with tolerance equal to the fixed desired error. Formula (22) gives the upper peaks. Every such sampler, when observed at ready boundaries, is itself an atomic transducer with at most its total number of labels. Hence the atomic lower bounds apply, irrespective of its internal timing. Taking binary logarithms and using (13) and (20) proves the stated leading coefficients. $\square$

This is not an exact finite-bit implementation of an irrational row at zero error. It is a fixed-error compiler with an explicitly charged input latch and bit counter. It does not make the row description or its construction free in an ordinary uniform-space model; those costs are outside both nonuniform measures. For the rational gates, exact rational atomic synthesis is additionally available, but exact rational sampling with unbounded rejection duration is a separate implementation choice.

8. OBJECTIVES, ANTECEDENTS, AND REMAINING DISTINCTIONS

8.1. Quantum and probabilistic automata. The prepare–rotate–query law is a numerical behavior, not just a language. Proposition 5.2 gives its exact quantum model and an explicit classical length-dependent obstruction. Chen–Wu’s revised paper [2] has a different title and sharper theorem than its April version: its stated cost is $n^2 + 1$ under strict cutpoints. Their companion work [3] treats the simulation cost of reversible quantum models through prepare–test and sign geometry. These compare threshold languages as quantum dimension varies. A shrinking positive acceptance margin can preserve such a language at all lengths and fail every fixed numerical tolerance in the present task.

The September symmetry study [4] likewise distinguishes reachable–observable rank from operational threshold memory. Its strict-cutpoint objective is not wordwise preservation of numerical probabilities. Our sphere example already has fixed real predictive rank four, so a rank theorem alone cannot yield its diverging compatible width. The present comparison does not claim that quantum/classical state separation itself is new.

Lumbreras–Ma–Thompson–Gu [5] analyze a stationary world model with an irrational phase, repeated histories, value error and decision loss, and supply an exact qutrit realization. Their Theorems 28–30 concern that stationary decision task. Here the classical machine may be redesigned for every finite horizon and may change its rows at every cut; one externally prescribed

terminal probability is tested. Neither a stationary peripheral-spectrum argument nor a strict-cutpoint conversion supplies our finite nonuniform width bounds.

8.2. Spectral and geometric inputs. Bourgain–Gamburd [1] supplies the full spectral-gap input for Theorem 5.1. We do not prove a new expansion theorem or a new sphere-covering law. The gap converts finite command length into a uniform Lipschitz mixing scale; the full-word contraction then converts that scale into a hidden-register cost. This distinguishes the finite alphabet from the all-rotation-input theorem retained in [10]. The gap constant is fixed but not evaluated numerically. Constants may deteriorate with dimension, the generating set and the error margin.

The lower bound uses a probability law on external commands, not an adaptive policy implemented by the simulator. In the spectral-gap class the test letters are independent. In the monomial class they are correlated within width-dependent packets. Berkes–Borda [6] study the harmonic and Diophantine behavior of the uncompressed circle walk; compression into an endpoint register is an additional operation.

The transducer is a probabilistic ordered read-once branching program with real readout. For a binary alphabet its usual Boolean-input matrix Fourier coefficient is the ordered product of the matrices $(T_{t,0} + (-1)^{s_t} T_{t,1})/2$. This algebraic identity is not the packet converse. The Fourier-growth results for regular and width-three programs [7, 8, 9] have their stated hypotheses and objectives. An arbitrary stochastic row need not preserve the uniform state law. Conversely, bounds valid for all deterministic tables can sometimes pass by convexity to randomized rows. Private row choices should not be mistaken for a persistent shared seed available to the decoder.

8.3. Inherited statements and preservation. The sharp bounded-type binary theorem, nonuniform occupation, actual terminal-error bounds, the angular coefficient $l^2 - 1$, and the separate compact-input result of [10] remain in the unaltered supporting article. The corrected full-word lemma supplies the advertised general packet implication without a product-sufficiency restriction. For a compact command table, a Borel convex choice can be made by triangulating a finite polytope and using the first containing simplex in a fixed order; affine barycentric coordinates on finitely many closed simplices are Borel. No such infinite table is used by the finite-alphabet results.

The accumulated local deficiency certificate from the preceding work is not the minimum actual terminal error. Given a fixed profile in the sphere case, the present proof directly yields

$$(23) \quad \mathcal{E}_N(K) \geq \frac{1}{2} \left[\frac{\rho}{\sqrt{d}} - \exp\{-c_d K^{-2/(d-1)} \lfloor N/B_K \rfloor\} \right]_+.$$

Indeed Lemma 2.3 holds for every candidate’s actual error, and the packet product bounds q_N above. This remains valid when intermediate approximation errors cancel. A sum of executable local garbling defects is only an upper certificate for a particular terminal error, by TV contraction; applying (23) to it does not turn that certificate into an intrinsic equality.

8.4. Conclusion. There are now two finite noncommuting classes with distinct quantitative laws. Spectral-gap actions on spheres have polynomial width exponent $(d-1)/2$, with the displayed logarithmic order gap. Resonant monomial actions have matched cube-root order and need not be detected by a character of the full group. Both conclusions allow arbitrary hidden states and layer-dependent stochastic rows. The explicit rational gates give a numerical-probability comparison with a single qubit, not a new strict-cutpoint state-cost theorem.

The logarithmic gap for spectral-gap alphabets, optimal finite constants, and classification of all irrational cyclic angles remain quantitative questions. General adaptive experimental design is a different task. The finite-coin corollary concerns a self-paced nonuniform sampler and does not

identify program-table space with label space. These distinctions leave the proved finite-alphabet results intact; they are not hypotheses silently replaced during implementation.

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